

Amazon E-Commerce Database Management System

DBMS-SELLER AND INVENTORY MANAGEMENT

1.DATABASE DESIGN

1.1 Seller Table

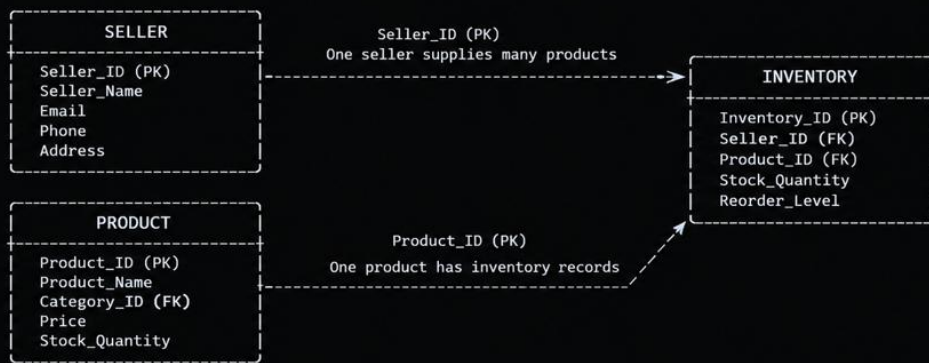
Seller_ID	Seller_Name	Email	Phone	Address
1	ABC Electronics	abc@gmail.com	9876543210	Chennai
2	Tech World	tech@gmail.com	9876543211	Coimbatore
3	Fashion Hub	fashion@gmail.com	9876543212	Madurai
4	Home Needs	home@gmail.com	9876543213	Chennai
5	Book Point	book@gmail.com	9876543214	Trichy

1.2 INVENTORY TABLE:

Inventory_ID	Seller_ID	Product_ID	Stock_Quantity	Reorder_Level
1	1	101	50	10
2	1	102	35	10
3	2	103	60	15
4	3	104	3	5
5	4	105	40	10

2.TABLE RELATIONSHIPS:

```
SQL> SELECT * FROM seller;
```



```
SQL> SELECT s.SELLER_ID, s.SELLER_NAME, i.INVENTORY_ID, i.PRODUCT_ID, i.STOCK_QUANTITY, i.REORDER_LEVEL
2 FROM SELLER s
3 JOIN INVENTORY i ON s.SELLER_ID = i.SELLER_ID;
```

SELLER_ID	SELLER_NAME	INVENTORY_ID	PRODUCT_ID	STOCK_QUANTITY	REORDER_LEVEL
1	ABC Electronics	1	101	50	10
1	ABC Electronics	2	102	35	10
2	Tech World	3	103	60	15
3	Fashion Hub	4	104	3	5
4	Home Needs	5	105	40	10

```
SQL> █
```

3.TABLE CREATION:

3.1 CREATE SELLER TABLE:

```
SQL> CREATE TABLE SELLER (
2 SELLER_ID NUMBER PRIMARY KEY,
3 SELLER_NAME VARCHAR2(100) NOT NULL,
4 EMAIL VARCHAR2(100) UNIQUE,
5 PHONE VARCHAR2(15),
6 ADDRESS VARCHAR2(200)
7 );
```

Table created.

3.2 CREATE INVENTORY TABLE:

```
SQL> CREATE TABLE INVENTORY (  
 2     INVENTORY_ID NUMBER PRIMARY KEY,  
 3     SELLER_ID NUMBER NOT NULL,  
 4     PRODUCT_ID NUMBER NOT NULL,  
 5     STOCK_QUANTITY NUMBER NOT NULL,  
 6     REORDER_LEVEL NUMBER  
 7 );
```

Table created.

4.SELLER DATA INSERTION:

```
SQL> INSERT INTO SELLER VALUES  
 2 (1, 'ABC Electronics', 'abc@gmail.com', '9876543210', 'Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES  
 2 (2, 'Tech World', 'tech@gmail.com', '9876543211', 'Coimbatore');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES  
 2 (3, 'Fashion Hub', 'fashion@gmail.com', '9876543212', 'Madurai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES  
 2 (4, 'Home Needs', 'home@gmail.com', '9876543213', 'Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES  
 2 (5, 'Book Point', 'book@gmail.com', '9876543214', 'Trichy');
```

1 row created.

5.INVENTORY DATA INSERTION:

```
SQL> INSERT INTO INVENTORY VALUES
  2 (1, 1, 101, 50, 10);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2 (2, 1, 102, 35, 10);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2 (3, 2, 103, 60, 15);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2 (4, 3, 104, 25, 5);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2 (5, 4, 105, 40, 10);

1 row created.
```

6.SELLER PRODUCT INFORMATION:

6.1 DISPLAY SELLER ANDDD PRODUCT DETAILS:

```
SQL> SELECT
  2     S.SELLER_ID,
  3     S.SELLER_NAME,
  4     I.PRODUCT_ID,
  5     I.STOCK_QUANTITY,
  6     I.REORDER_LEVEL
  7 FROM SELLER S
  8 JOIN INVENTORY I
  9 ON S.SELLER_ID = I.SELLER_ID;

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID STOCK_QUANTITY REORDER_LEVEL
-----
          1
ABC Electronics
    101             50             10

          1
ABC Electronics
    102             35             10

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID STOCK_QUANTITY REORDER_LEVEL
-----
          2
Tech World
    103             60             15
```

7.INVENTORY STATUS:

7.1 DISPLAY AVAILABLE PRODUCTS:

```
SQL> SELECT
  2     I.PRODUCT_ID,
  3     I.STOCK_QUANTITY,
  4     I.REORDER_LEVEL
  5 FROM INVENTORY I
  6 WHERE I.STOCK_QUANTITY > 0;

PRODUCT_ID STOCK_QUANTITY REORDER_LEVEL
-----
          101             50             10
          102             35             10
          103             60             15
          104              3              5
          105             40             10

SQL>
```

7.2 DISPLAY UNAVAILABLE PRODUCTS:

```
SQL> SELECT
2      I.PRODUCT_ID,
3      I.STOCK_QUANTITY,
4      I.REORDER_LEVEL
5  FROM INVENTORY I
6  WHERE I.STOCK_QUANTITY = 0;
```

8.INVENTORY UPDATE:

Demonstrate update stock information

```
SQL> SELECT
2      S.SELLER_NAME,
3      I.PRODUCT_ID,
4      I.STOCK_QUANTITY,
5      I.REORDER_LEVEL
6  FROM SELLER S
7  JOIN INVENTORY I
8  ON S.SELLER_ID = I.SELLER_ID;
```

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY REORDER_LEVEL

ABC Electronics	101	50	10
-----------------	-----	----	----

ABC Electronics	102	35	10
-----------------	-----	----	----

Tech World	103	60	15
------------	-----	----	----

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY REORDER_LEVEL

Fashion Hub	104	25	5
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Home Needs	105	40	10
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